

AdaptIQ – Minimum Viable Product (MVP)

Version: 1.0 (Hackathon Edition)

Project Type: AI-Powered Adaptive Learning Interface

1. Problem Statement

Over 300 million learners worldwide struggle with standardized educational content that does not adapt to different cognitive processing styles. Traditional learning materials are designed for neurotypical learners and do not account for differences in attention span, reading comfort, or logical step processing. AdaptIQ solves this by restructuring content presentation to match different cognitive learning modes.

2. MVP Objective

The MVP aims to build a working prototype that demonstrates real-time AI-driven content restructuring into multiple adaptive learning modes using a local AI model (Ollama – Llama3 8B). The focus is on reliability, structured output, and visual clarity for hackathon demonstration.

3. Core Features (MVP Scope)

- Content Input Panel: Users paste educational material (max 5000 characters).
- Multi-Mode Content Adaptation: AI generates three structured modes in a single request: Read-Easy Mode, Focus Mode, Step-by-Step Mode.
- Focus Mode Progressive Reveal: Sections are displayed one at a time using JavaScript pacing control.
- AI-Generated Visual Diagram (Focus Mode): Mermaid diagram generated dynamically to explain core concept.
- Mode Switching: Users can switch between modes instantly without additional AI calls.
- Quiz Generation: AI generates 3 multiple-choice questions based on adapted content.
- Text-to-Speech (TTS): Browser-native Web Speech API reads adapted content aloud.
- Session Caching: Adapted modes stored in Flask session to avoid repeated model calls.

4. System Architecture (MVP)

Frontend: HTML, CSS, JavaScript, Bootstrap 5

Backend: Python Flask Framework

AI Engine: Ollama (Llama3 8B Instruct)

Diagram Rendering: Mermaid.js

Audio: Web Speech API

Storage: Flask Session (No database in MVP)

5. Cognitive Modes Description

Read-Easy Mode

Short paragraphs (max 3 sentences), simplified vocabulary, increased line spacing, and highlighted key terms to reduce visual stress.

Focus Mode

Content divided into labeled sections (max 100 words each), progressive reveal mechanism, recap lines, and a dynamically generated visual diagram for conceptual clarity.

Step-by-Step Mode

Logical breakdown of complex ideas into ordered steps, each presented in isolated cards. Formulas are translated into plain English explanations.

6. Functional Flow (Adaptation Process)

1. User pastes content and clicks 'Adapt Content'.
2. Frontend sends POST request to Flask backend.
3. Backend applies cognitive rule constraints.
4. Structured prompt sent to Ollama.
5. Model returns structured output with markers.
6. Backend parses and validates response.
7. All modes cached in session.
8. Frontend renders selected mode.

7. Non-Functional Requirements (MVP)

- Response time under 10–15 seconds per adaptation.
- Runs fully offline using local AI model.
- No database dependency.
- Error handling for malformed AI output.
- Clean UI with no console errors during demo.

8. Out of Scope (Post-MVP)

- User authentication and accounts.
- Cloud deployment scaling.
- Multi-language support.
- Analytics dashboard.
- Teacher/admin panel.